James Ensminger

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EXPERIENCE

Applications Engineer Nov 2023 - Present

PowerFlex | Mountain View, CA and Los Angeles, CA

- Enhanced EV charger reliability through bench testing, innovative tooling, and data-driven R&D.
- Advanced distributed control system design iterations with the HW team for optimized power distribution, networking, and telemetry between onsite network devices and DERs (reduced site downtime across North America by 22%).
- Led the development and launch of customer-facing and internal-facing technical documentation platforms on PowerFlex HW and SW products and services.

Asset Health and Performance Center Engineering Intern Jun 2021 - Sep 2021

Pacific Gas and Electric | San Francisco, CA

- Leveraged GIS to provide informative data on future EFD (Early Fault Detection) and DFA (Distribution Fault Anticipation) line sensor technology installments across Northern California's electric grid.
- Managed and proactively coordinated with multiple teams to compile a quarterly voltage curtailment report on all solar powered line sensor units, while taking initiative to reduce data compilation processing time by 20% (saving 1 hour) through root cause analysis and process improvement methodologies.

PROJECTS AND ACHIEVEMENTS

3x3 Systolic Array Mar 2026 - May 2026

- Fully custom design in 45nm CMOS for accelerating matrix multiplication by enabling parallel computation across 9 synchronous PEs (processing elements) each performing MAC (multiply-accumulate) operations.
- Achieved 385MHz maximum operating frequency, 450μW average power, and 21,730μm² layout footprint with clean DRC/LVS sign-off.

Multi-Device Wireless Charger Dec 2022 - Jun 2023

- Created a Qi standard wireless charger with increased freedom of placement for multi-device charging, an intuitive visual charge indicator determined by stakeholders, and an efficient heat management system enabling it to operate 5°C cooler than chargers on the market.

Ground Bounce PCB Test Circuit Mar 2022 - Jun 2022

- Designed a PCB test circuit to experimentally observe the electrical phenomena of ground bounce.
- Developed the PCB design with OrCAD Capture and Allegro PCB Editor, then milled it using an M60 LPKF.

EDUCATION

University of Southern California | Master of Science in Electrical Engineering Aug 2025 - Present

- **Involvement:** Flexible Electronics research at USC Center for Advanced Manufacturing, Viterbi Graduate Orientation Leader
- **Coursework:** Modern Solid-State Devices, Power Systems, MOS VLSI Design, Nanotechnology, Computing Principles, VLSI System Design

University of California, Santa Cruz | Bachelor of Science in Electrical Engineering Sep 2019 - Jun 2023

- **Clubs:** MEP, NeuroTechSC, SlugSat, UCSC IEEE
- **Grader:** ECE178 and ECE224 Semiconductor Device Electronics [Winter Quarter 2023]

SKILLS, TOOLS, AND TECHNOLOGIES

- **Electrical:** OrCAD Capture, PSpice, Allegro PCB Editor, Altium Designer, Cadence Virtuoso, Ansys Maxwell, Hardware Development/Debugging, Microcontrollers, RTL Design, FPGA, PLC programming (IEC 61131-3), Modbus TCP, Computer Networking, Soldering, Lab Equipment (Oscilloscopes, Signal Generators, Power Supplies, DMMs, Spectrum Analyzers)
- **Software:** Verilog, Python, C/C++, Java, SQL, PowerShell, Linux, Git, MATLAB
- **Mechanical:** SolidWorks, 3D Printing, Ansys Mechanical